From here and there

- Intel, together with collaborators, recently created the first silicon qubits at scale at its D1 manufacturing factory in Hillsboro, Oregon. The result is a process that can fabricate more than 10,000 arrays with several silicon-spin qubits on a single wafer with greater than 95% yield. The qubit count and yield are much higher than the typical university and laboratory processes used today.
- Scientists at the Harvard-MIT Center for Ultracold Atoms have demonstrated a programmable quantum simulator capable of operating with 256 qubits. The system uses arrays of highly-focused laser beams, to trap individual qubits and drag them into desirable arrangements to perform meaningful calculations. This will help to build large-scale, practical quantum computers with great applicability, in the near future.

cryptographic key generation platform based on verifiable quantum randomness. They are also developing solutions for chemistry, natural language processing, machine learning, and more.

IonQ, the first quantum computing startup to go public, uses individual atoms as their qubits. According to company literature, "At IonQ, we take a different approach, and use a naturally occurring quantum system: individual atoms. These atoms are the heart of our quantum processing units. We trap them in 3D space, and then use lasers to do everything from initial preparation to final readout."

Despite all the advancements, Prof. Dumke feels that, "Hardware is and will be always the bottleneck. This is true for any technology, and for classical computers as well as for quantum computers. The reason why we do not have a universal quantum computer yet is that the hardware is not mature enough. I can foresee that specific hardware solutions made from noisy processors will be tailored to solve special use-cases in the near future."

Making quantum more accessible

More than pure quantum tech advancements, what is more interesting today are the different approaches being taken to make quantum computing accessible to more people. Two of the most significant efforts in this direction involve offering hybrid solutions, which combine quantum computing with classical computing setups; and full-stack cloud solutions that mask the complexity and let more people tap the benefits of quantum.

Liscouski says, "Our goal from the beginning has been to make quantum computing as available as possible so that business users with no quantum expertise can benefit from it. That's why we invented Qatalyst, a ready-to-run quantum optimisation and machine learning software that masks the massive complexity and effort needed to solve even basic problems on quantum computers."

Through QUBT U, they also provide the Qatalyst software to students, allowing them access to the world of quantum computing without having to be quantum programmers! "We believe that workforce development is a key component of our responsibility to our society," says Liscouski.

IBM's plans to build a 4,000+ qubit processor by 2023 goes hand-in-hand with significant milestones to build an intelligent quantum software orchestration platform that will abstract away the noise and complexity of quantum machines, and allow large and complicated problems to be easily broken apart and solved across a network of quantum and classical systems. "There is an inherent strong coupling of quantum and classical computers. Both paradigms of computing—classical and quantum—will work seamlessly together to solve different parts of complex computing problems that are best suited for their respective strengths and capabilities," says Subramaniam.

IBM is integrating quantum computing with high-performance computing and hybrid cloud technologies in serverless implementations that remove the complexity of infrastructure management, putting the focus on coding only.

"It took 60 years to abstract software in classical computing to the point where users could input a simple line of code into a templated program in order to build an app or website. Quantum computing is going through a similar process—within the decade. IBM released the Qiskit Optimization Module 2020 as a first step in making frictionless quantum computing a reality by 2025.

"In the future, a program will be handed off to the cloud and vast quantum and classical resources will be employed, and within the blink of an eye the solution will be returned perfectly optimised. The Qiskit Optimization Module enables easy, efficient modelling of optimisation problems for developers and optimisation experts without quantum expertise. It uses classical optimisation best practices and masks complex quantum programming," says Subramaniam.

IBM has installed on-site quantum systems in Germany and Japan, with plans for installations in Korea, Canada, and in the USA at the Cleveland Clinic. They also have over 20 systems in their Poughkeepsie and Yorktown locations in the USA, which can be accessed over the cloud.

In 2016, IBM became the first company to put quantum computers on the cloud. This was a turning point. "Since then, we have actively built up an active community of more than 400,000 users and 190+ organisations in the IBM Quantum Network that are running more than 4 billion circuits every day and exploring practical applications to realise the wide-ranging benefits of this technology for business and society," says Subramaniam. From India, IIT Madras became the first institution to